

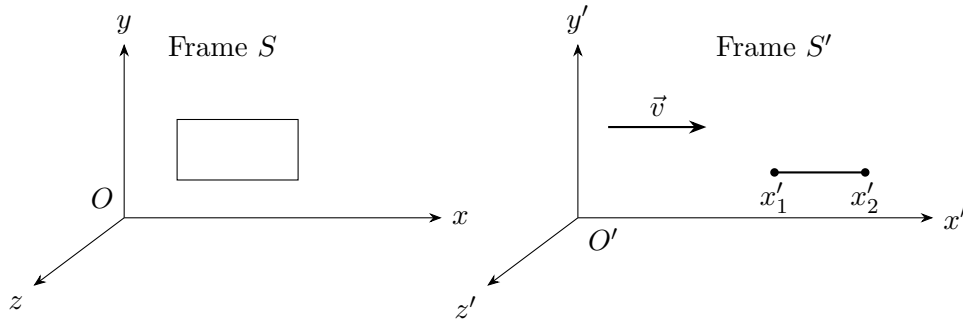
Special Relativity Notes

Lorentz Transformation, Length Contraction, Time Dilation and Mass Variation

21/09/26

Lorentz Transformation of Coordinates Parallel to the x -axis

Frame S' (the flight) moves with velocity $\vec{v}$ along the x -axis relative to frame S (outside observer).



Lorentz transformation

$$x' = \frac{x - vt}{\sqrt{1 - \frac{v^2}{c^2}}}, \quad y' = y, \quad z' = z, \quad t' = \frac{t - \frac{xv}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

1 Length Contraction using L.T.

Q. What will be the length of this rod measured by the observer sitting outside the flight?
(Thought experiment) $L = ?$

The observer sitting *inside* the flight measures the length of the rod as

$$L_0 = x'_2 - x'_1.$$

Here t' is the same for measuring x'_1 and x'_2 .

Approach (I)

$$\begin{aligned}
 L_0 &= x'_2 - x'_1 \\
 &= \frac{x_2 - vt_2}{\sqrt{1 - \frac{v^2}{c^2}}} - \frac{x_1 - vt_1}{\sqrt{1 - \frac{v^2}{c^2}}} \\
 &= \frac{x_2 - x_1}{\sqrt{1 - \frac{v^2}{c^2}}} + \frac{vt_1 - vt_2}{\sqrt{1 - \frac{v^2}{c^2}}}
 \end{aligned}$$

Now impose $t'_1 = t'_2$:

$$\begin{aligned}
 t_1 - \frac{x_1 v}{c^2} &= t_2 - \frac{x_2 v}{c^2} \\
 t_1 - t_2 &= \frac{vx_1}{c^2} - \frac{vx_2}{c^2} = \frac{v}{c^2}(x_1 - x_2) \\
 v(t_1 - t_2) &= \frac{v^2}{c^2}(x_1 - x_2)
 \end{aligned}$$

Substituting this into the expression for L_0 :

$$\begin{aligned}
 L_0 &= \frac{x_2 - x_1}{\sqrt{1 - \frac{v^2}{c^2}}} + \frac{\frac{v^2}{c^2}(x_1 - x_2)}{\sqrt{1 - \frac{v^2}{c^2}}} \\
 &= \frac{x_2 - x_1}{\sqrt{1 - \frac{v^2}{c^2}}} - \frac{(x_2 - x_1)\frac{v^2}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \\
 &= (x_2 - x_1) \frac{1 - \frac{v^2}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \\
 L_0 &= L \sqrt{1 - \frac{v^2}{c^2}}
 \end{aligned}$$

× **Wrong**

Approach (II)

Here the outside observer marks both ends at the same instant t (so $t_1 = t_2 = t$):

$$\begin{aligned}
 L_0 &= x'_2 - x'_1 \\
 &= \frac{x_2 - vt}{\sqrt{1 - \frac{v^2}{c^2}}} - \frac{x_1 - vt}{\sqrt{1 - \frac{v^2}{c^2}}} \\
 L_0 &= \frac{x_2 - x_1}{\sqrt{1 - \frac{v^2}{c^2}}}
 \end{aligned}$$

$$L = L_0 \sqrt{1 - \frac{v^2}{c^2}}$$

with $L = x_2 - x_1$

✓ **Correct**

Approach (III)

Start from $L = x_2 - x_1$ and use the inverse transformation with t' the same for both ends:

$$\begin{aligned} L &= x_2 - x_1 \\ &= \frac{x'_2 + vt'}{\sqrt{1 - \frac{v^2}{c^2}}} - \frac{x'_1 + vt'}{\sqrt{1 - \frac{v^2}{c^2}}} \\ &= \frac{x'_2 - x'_1}{\sqrt{1 - \frac{v^2}{c^2}}} \\ L &= \frac{L_0}{\sqrt{1 - \frac{v^2}{c^2}}} \end{aligned}$$

✗ **Wrong**

2 Time Dilation using L.T.

At the location x' , the two events happen at t'_1 and t'_2 , so

$$T_0 = t'_2 - t'_1 \quad (\text{property time}).$$

T is the time interval measured by the observer outside the flight:

$$T = t_2 - t_1.$$

Attempt 1

$$\begin{aligned} T_0 = t'_2 - t'_1 &= \frac{t_2 - \frac{vx_2}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} - \frac{t_1 - \frac{vx_1}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \\ &= ? \end{aligned}$$

✗ **Wrong** ($\because x_1 \neq x_2$, so the position terms do not cancel)

Attempt 2

$$\begin{aligned}
 T = t_2 - t_1 &= \frac{t'_2 + \frac{vx'_2}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} - \frac{t'_1 + \frac{vx'_1}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \\
 &= \frac{t'_2 - t'_1}{\sqrt{1 - \frac{v^2}{c^2}}} \quad (\because x'_1 = x'_2, x' \text{ is fixed})
 \end{aligned}$$

$$T = \frac{T_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

✓ Correct

Why some approaches fail

In length contraction, the measurement is correct only when both endpoints are marked at the exact same instant in the outside observer's frame ($t_1 = t_2$). Setting $t'_1 = t'_2$ means the marks were simultaneous only inside the moving frame, so the outside observer recorded the ends at different times as the rod moved past.

Similarly, in time dilation the two events occur at the same place in the moving frame, $x'_1 = x'_2$, but at different places in the outside frame, $x_1 \neq x_2$.

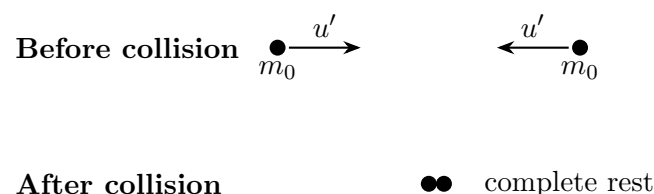
3 Variation of Mass with Velocity of a Frame

Q. How does the mass of an object change with respect to the velocity of a frame?

Thought experiment

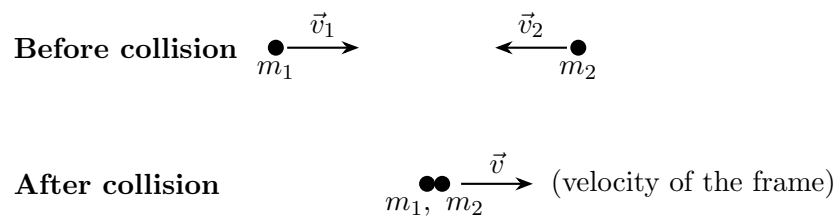
Frame S' (the flight) moves with velocity v along x relative to frame S .

As described by the observer inside the flight (S')



Description given by the observer outside the flight (S)

Q. What will be the description as given by the observer outside the flight?



Approach

We have to use conservation of momentum in both frames and find how m_1 and m_2 are related to each other with respect to the frame velocity.

Conservation of momentum (frame S)

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v$$